WEATHER

Read your own barometer

Version 1.0 — reasoning from the observations you took, with no network at all. It cannot forecast for you. It can make you a forecaster.

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1. The honest limit, first

With no network this cannot forecast. No GRIBs, no synoptic charts, no models, and there never will be any: there is no connection here to make one with.

What it does instead is reason from the **observation** column — ground truth you measured with your own instruments — which is the one category of weather data that is never wrong, and the one still available when the antenna comes down.

> Celestial navigation is what you do when the GPS dies. This is what you do when the sat comms die.

Everything below follows from that.

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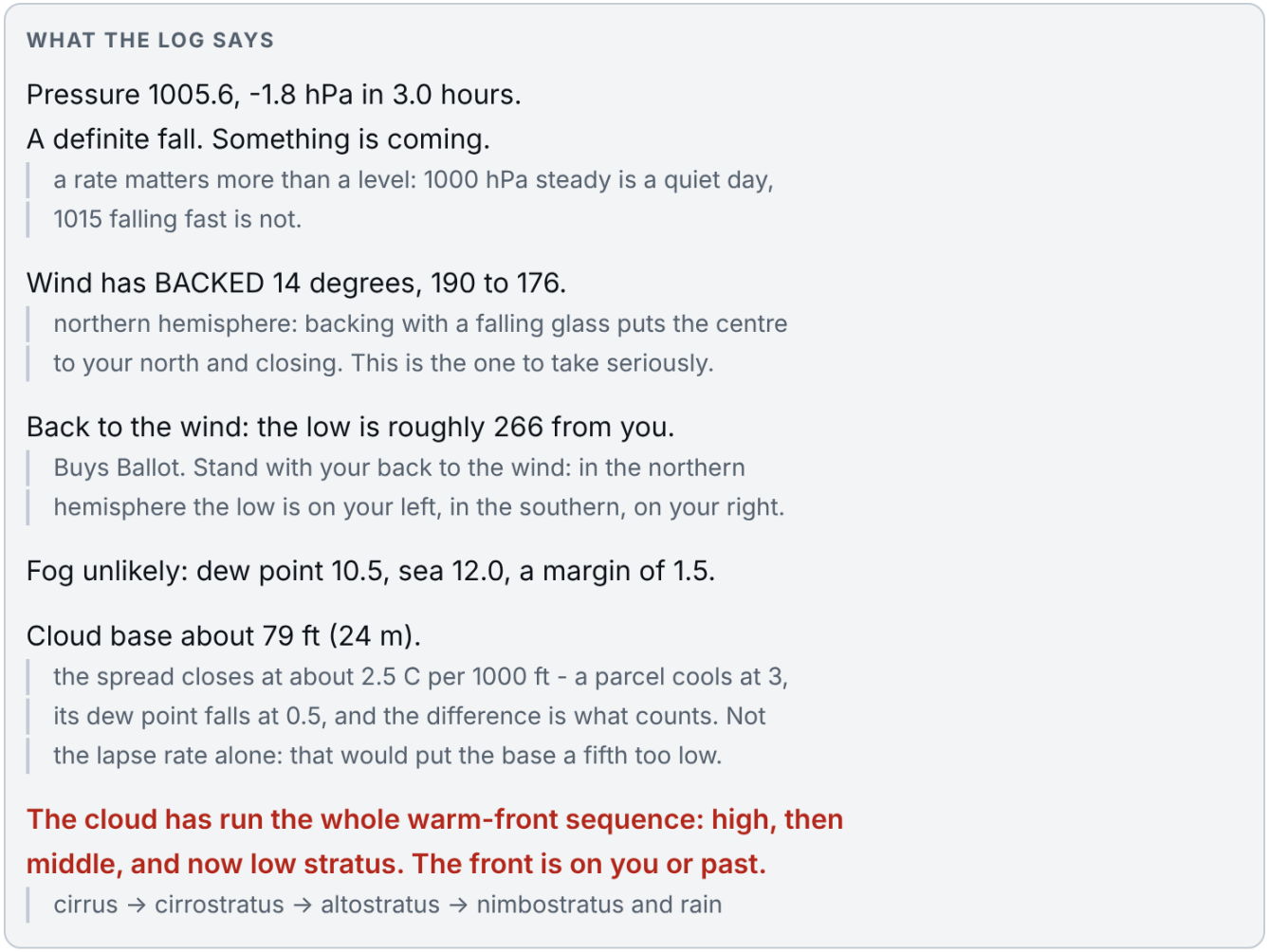
2. Where its numbers come from

It reads the log that `deck-log` writes — the same records, in the same store, on the same device. Nothing is copied between the two tools and there is nothing to keep in step.

So the quality of what this tool says is the quality of your observations. Pressure to a tenth of a millibar, the cloud by level, the dew point, and the swell period are what the reasoning actually uses. A log of position and course alone gives it almost nothing to work with.

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3. What the log says



▸ Show it as the terminal does

Figure 1 — the reasoning, with every step shown and its rule underneath.

`weather what` (or **What the log says**) takes your last observations and reasons out loud. Every line is followed by the rule it came from, in smaller type, because the point is that you can do this yourself afterwards.

The pressure

The rate matters more than the level. 1000 hPa steady is a quiet day; 1015 falling fast is not. The wording follows the three-hour change:

3.0 hPa or more of fall in 3 h	*a serious fall — expect a deepening system and a rising wind*
1.5 to 3.0	*a definite fall. Something is coming*
within 1.5 either way	*near steady*
1.5 or more of rise	*rising — the gradient is easing or a ridge is building*

It corrects for the atmospheric tide first. The atmosphere has its own semidiurnal pressure wave, largest in the tropics, and near the equator it is big enough to hide a real

trend or invent one. The correction needs your longitude, which is why `weather lon <deg>` exists.

The wind

Backed or veered, against the previous observation, with Buys Ballot applied: stand with your back to the wind and in the northern hemisphere the low is on your left. The tool gives the bearing of the centre from you.

Backing with a falling glass is the one to take seriously — it puts the centre to your north and closing.

Fog, and the cloud base

1. **Fog**: the margin between dew point and sea temperature. When the air's dew point is at or above the sea temperature, you are in the business of making fog.
2. **Cloud base**: the spread between temperature and dew point closes at about 2.5 °C per 1000 ft — a parcel cools at 3 and its dew point falls at 0.5, and the *difference* is what counts. Using the lapse rate alone puts the base about a fifth too low.

The cloud sequence

The classic warm front is cirrus, then cirrostratus, then altostratus, then nimbostratus and rain. The tool watches your `ch` / `cm` / `cl` codes over successive observations and tells you how far through that sequence you are — which is a judgement you cannot make from a single observation, and the reason the cloud is logged by level rather than as "overcast".

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4. The forecast loop, and why yours comes first

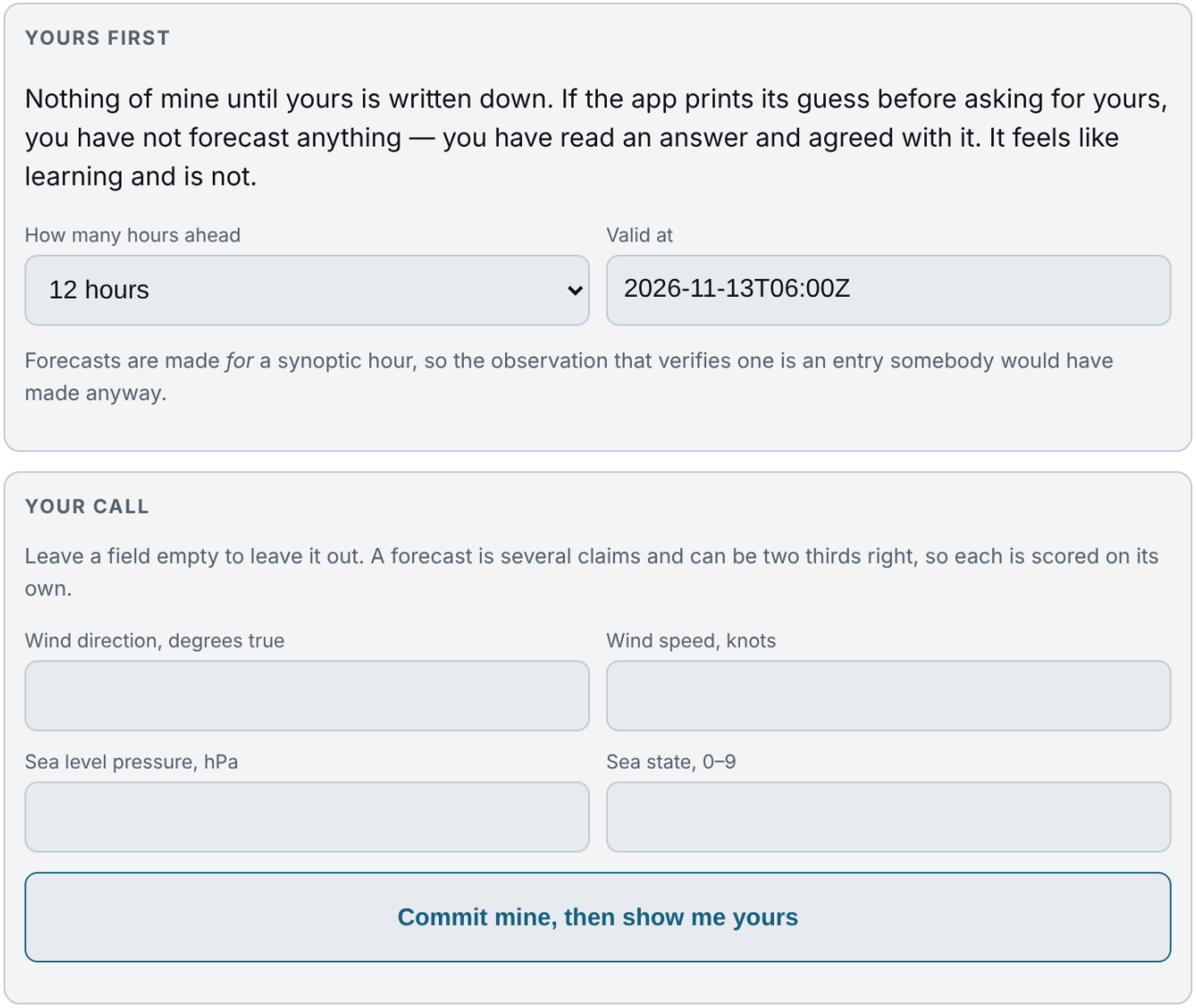


Figure 2 — yours first. Nothing of the machine's until yours is written down.

Nothing of the machine's appears until your own forecast is recorded. If the app prints its guess first, you have not forecast anything: you have agreed with something. The loop is deliberate:

1. You write your forecast — wind direction and speed, pressure, and the weather.
2. Only then does the tool show its own two: the **rules** forecaster, and **persistence** (tomorrow is like today).
3. When the verifying observation arrives, all three are scored against it.

Persistence is in there because it is the honest benchmark. It is surprisingly hard to beat over 12 hours, and a forecaster who cannot beat it is not yet adding anything.

5. The score

THE SCORE

Weighted error points, from the WxChallenge: half a point per knot, one per millibar, a tenth per degree of wind direction — so a near miss scores like a near miss. **Lower is better.**

No forecast has reached its valid time yet, or no observation has been logged near one. Make a forecast, then log the observation it was for, and this fills in.

WHY PERSISTENCE IS ON THE SHEET

"In twelve hours it will be much as it is now" is the honest floor. It is hard to beat at short range, and a forecast that does not beat it has shown no skill at all. Standard practice in operational meteorology, and free to compute.

Scoring the rules as well as you keeps the app honest — its misses go on the record — and teaches calibration: a rule right seven times in ten is useful *once you know it is seven in ten*. And it lets you eventually win, which is the point. A training tool that cannot be outgrown is badly built.

▸ Show it as the terminal does

Figure 3 — you, the rules, and persistence, scored against what actually happened.

Weighted error points, from the WxChallenge convention: half a point per knot of wind speed, one per millibar, a tenth per degree of wind direction — so a near miss costs little and a bad miss costs a lot.

The score exists to answer one question honestly: **are you getting better, and are you better than persistence?**

6. The 500 millibar chart

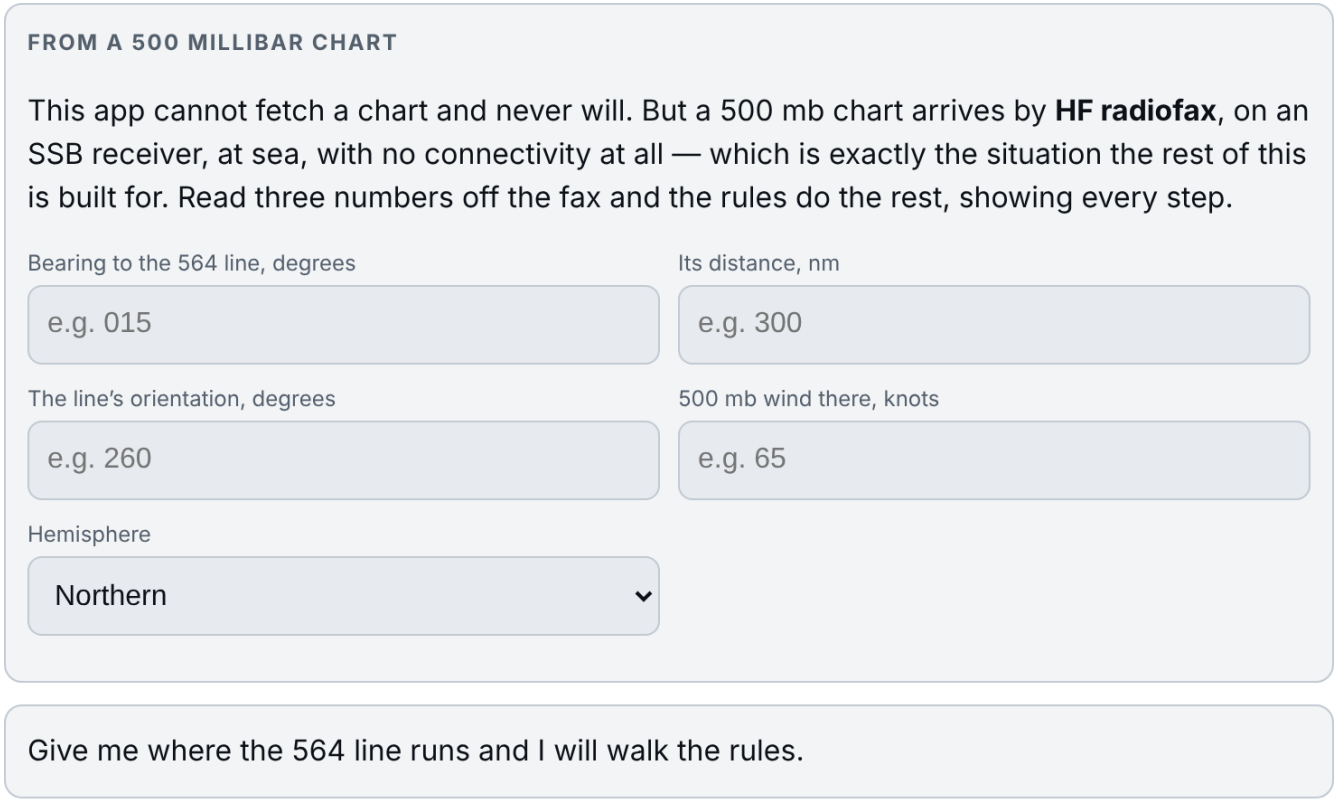


Figure 4 — reasoning over a chart that arrived by radiofax.

This tool cannot fetch a chart and never will. But a 500 mb chart does arrive at sea, by HF radiofax on an SSB receiver, with no internet involved at all — and it is the one upper-air product worth having, because the 500 mb flow is what steers the surface systems.

So `weather chart` asks for what you can read off the fax with a ruler, and the rules do the rest, showing every step:

Bearing to the **564 line**	the 5,640 m contour: the storm track runs along it
Its distance, in miles	
The line's orientation	which way it lies across your position
The 500 mb wind there, in knots	how fast the pattern is being steered
Hemisphere	which side of the line is the poleward one

The first question it answers is whether the 564 line is **poleward of you**, because equatorward of it is where you want to be. The rules are from Sienkiewicz and Chesneau, *Mariner's Guide to the 500-Millibar Chart* (NOAA, public domain), and are listed in `docs/SOURCES.md`.

7. The lessons

LESSONS

Each one earns its place by changing what the app says or what you would do. One of them says plainly where that stops being true, which is the point of the rule.

- **Air is a fluid, and moist air is lighter**
- **The three kinds of weather data, and which one you have**
- **The atmospheric tide: why the glass falls every afternoon**
- **Coriolis, which is not a force**
- **Gradient wind, surface wind, and friction**
- **Lapse rates, stability, and where the cloud base is**
- **The heat engine, and the three cells**
- **The 500 millibar chart and the 564 line**
- **How a tropical cyclone is made, and how to keep away**
- **The tilt, the seasons, and an honest boundary**

Figure 5 — ten lessons, each one earning its place.

Each lesson changes either what the tool says or what you would do. One of them says plainly where the reasoning stops being reliable, which is the most important of the ten.

`weather learn` lists them; `weather learn <key>` opens one.

8. Commands

`weather`	the menu
`weather what`	what your log says is coming, with the reasoning
`weather learn [key]`	a lesson; no key lists them
`weather chart`	reason over a 500 mb radiofax chart
`weather forecast`	yours first, then mine, then both are scored
`weather score`	how you, the rules and persistence are doing
`weather lon <deg>`	your longitude, east positive — for the pressure tide
`weather day` / `night` / `plain`	colour mode
`weather about` / `version`	

It reads the deck log; `deck-log` is what writes it. It never forecasts from a model, because there is no network here and never will be.

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9. The display modes

The display: day, dark and night

The browser build has three display modes. They are not decoration: one of them exists to protect your night vision, and choosing the wrong one at the wrong moment costs you something real.

Mode	What it is for
NIGHT	Red on black, for a dark-adapted eye. The default.
DARK	Near-black with light text, for below decks, dusk, or indoors.
DAY	White with near-black text, for daylight and for printing.

Why night is the default

A tool you open on deck in the dark must not start bright. A bright panel costs a dark-adapted eye roughly **twenty minutes** to recover from, and twenty minutes is a long time to be unable to see a mark or a ship’s lights properly. So the first time you open any of these tools you get night, and from then on you get whatever you last chose.

Three buttons, not a cycle

The three modes are three separate buttons, and the one you are in is shown as selected. That is deliberate.

1. **A cycle can take you somewhere you did not intend.** One stray press with a cycling control can land you on DAY at three in the morning.

2. **A cycle cannot tell you where you are**, only what the next press would do. Three buttons show the current mode at a glance without changing it.
3. **Going darker is always one tap.** DARK and NIGHT are immediate from anywhere, because going darker never costs you anything.
4. **Going from NIGHT to DAY takes a deliberate half-second press.** Hold the DAY button and it fills as it goes; let go early and nothing happens. Nothing bright appears on the screen until the change actually completes. That one transition is guarded because that one transition is the only one that costs you your dark adaptation.

The setting follows you

The mode is remembered for all of the tools together, and for the course pages that embed them. Set night in one and you are in night in the next. It is remembered in the browser, so it survives closing the app.

It is right before the page draws

The mode is set in the first instructions the page runs, before any of it is painted. This matters on a first visit over a poor connection: set it later and the page would appear in one palette and then jump to another, which at night is the difference between keeping your night vision and losing twenty minutes of it. If nothing has been stored yet, or whatever was stored cannot be read, the page opens in **night**.

The controls follow the palette too

Sliders, scroll bars and the other controls the browser draws are told which world they are in, so they are drawn in the mode you chose rather than the browser's own light default. Without that, a night screen can carry a bright white slider across the middle of it.

If you use High Contrast Mode

On Windows, High Contrast replaces every colour on the page with the ones your system theme specifies. That applies here too: the tools do not override it, because it is your setting and you chose it.

Nothing is lost by it. Where a colour carries meaning — the navigation lamps in colregs — the letter beside it says the same thing, so a lamp is still identifiable when the hue has gone.

One thing to know if you need both. High Contrast wins over night mode. If you need night vision on deck *and* high contrast to read, choose a dark high-contrast theme in Windows; the tools will follow it.

What the display can and cannot do in sunlight

Worth knowing before you blame the app on a bright day.

In sunlight, light reflecting off the glass lands on the dark parts of the screen and the light parts alike. That lifts everything together and flattens the difference between them, so the contrast you actually see collapses no matter what colours the page uses:

Condition	Body text, as seen
Indoors	about 10:1
Overcast on deck	about 6:1
Bright	about 3.5:1
Full sun on the glass	about 2.3:1

At the extreme, every colour scheme converges. Pure black on pure white measures the same 2.3:1 in full sun as the day mode does, because the glare swamps the difference. A different palette is not the answer, and nor is a different shade of white: across every plausible choice of background the contrast changes by under 4%, which is below what an eye can see.

What does work, in order:

1. **Turn the screen brightness up.** This is the one large lever, and it is the device’s, not the app’s.
2. **Shade the screen** with a hand, your body or the sprayhood. Removing the reflection is worth more than any change the app can make.
3. **Use DAY mode**, which at least puts the maximum luminance the panel can give behind the darkest text it can draw.

A note for printing

Documents and their figures are made in **day** mode. A screen captured in dark or night mode prints as a solid block of toner and loses the fine type in a working, so the printed pages show the day display even if that is not the one you use at sea.

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10. Cautions

1. **This is not a forecast service.** It reasons from what you logged. It has no model, no satellite and no warning of anything over the horizon that has not announced itself in your own observations.
2. **It cannot see a system that is not yet showing.** A squall line fifty miles away affects nothing you have measured until it does.
3. **Garbage in.** An instrument that is out, a pressure read to the nearest millibar, or cloud logged as “cloudy” will give confident reasoning from bad input. The tool says what it is reasoning from; check that first when an answer surprises you.
4. **Near the equator, log your longitude.** Without it the atmospheric tide correction cannot be made, and the semidiurnal wave there is large enough to look like a trend.
5. **Where this and an official forecast differ, take the official one** and treat the difference as something to understand rather than to argue with.